

TOORA-KHEM, Russian Federation

WMO: 361030

Lat: 52.4703N

Lon: 96.1192E

Elev: 920

StdP: 90.76

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.4	-38.8	-43.2	0.1	-40.9	-40.9	0.1	-38.6	5.7	-16.1	4.4	-17.1	0.0	340	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	28.2	16.7	26.1	15.7	24.2	15.3	18.1	24.8	17.2	23.6	16.2	22.4	2.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.8	12.5	20.5	14.8	11.7	19.5	13.9	11.1	18.7	55.2	24.5	52.0	23.4	49.1	22.6	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.7	4.5	3.7	DB	-44.7	31.4	3.0	2.2	-46.9	33.0	-48.6	34.3	-50.3	35.5	-52.5	37.1	(4)
(5)			WB	-43.9	20.4	2.4	1.7	-45.6	21.6	-47.0	22.5	-48.3	23.5	-50.0	24.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.8	-25.0	-18.8	-8.4	0.8	6.8	13.2	15.5	13.7	6.7	-2.0	-14.8	-22.1	(6)	
	DBStd	14.93	5.54	6.04	6.26	4.63	4.09	4.12	2.86	3.16	4.06	5.26	7.30	6.61	(7)	
	HDD10.0	5103	1086	805	570	278	117	17	2	6	113	372	744	993	(8)	
	HDD18.3	7719	1344	1038	828	527	358	159	92	147	349	631	994	1252	(9)	
	CDD10.0	438	0	0	0	1	18	112	173	119	14	0	0	0	(10)	
	CDD18.3	13	0	0	0	0	0	4	6	3	0	0	0	0	(11)	
	CDH23.3	598	0	0	0	0	39	232	211	112	5	0	0	0	(12)	
	CDH26.7	138	0	0	0	0	6	61	49	23	0	0	0	0	(13)	
	Wind	WSAvg	1.1	0.4	0.6	1.1	1.7	1.8	1.4	1.1	1.1	1.0	0.8	0.5	(14)	
(15) Precipitation	PrecAvg	334	10	6	6	14	30	52	68	68	38	15	14	14	(15)	
	PrecMax	483	18	14	14	28	63	90	134	121	83	37	31	24	(16)	
	PrecMin	202	3	1	1	3	4	18	17	21	9	3	4	6	(17)	
	PrecStd	65	4	3	3	5	14	19	29	26	19	8	7	5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.5	0.6	11.3	20.4	27.1	31.0	30.9	29.3	23.7	15.0	2.5	-3.2	(19)	
		MCWB	-8.4	-2.4	4.4	9.2	13.3	17.4	18.4	17.6	14.0	7.9	-0.2	-3.9	(20)	
	2%	DB	-10.8	-2.9	8.0	16.5	24.0	28.6	27.8	26.1	20.7	11.3	-0.1	-6.5	(21)	
		MCWB	-11.4	-4.6	2.3	7.2	11.9	16.3	16.9	16.8	12.6	6.2	-1.4	-7.2	(22)	
	5%	DB	-13.8	-6.0	5.3	13.3	20.8	26.1	25.6	24.0	17.9	8.1	-1.9	-9.1	(23)	
		MCWB	-14.4	-7.6	0.9	5.8	10.4	15.3	16.3	16.2	11.6	4.1	-3.4	-9.7	(24)	
	10%	DB	-16.1	-8.9	2.3	10.2	17.6	23.8	23.5	21.6	15.1	5.4	-4.0	-11.8	(25)	
		MCWB	-16.6	-10.2	-1.1	4.4	9.0	14.6	16.0	14.7	9.9	2.5	-5.2	-12.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.3	-1.9	4.5	10.1	14.4	19.1	19.6	19.0	14.9	8.6	0.3	-3.9	(27)	
		MCDB	-7.6	0.3	10.6	17.9	26.0	27.8	27.7	26.4	21.7	13.3	1.3	-2.9	(28)	
	2%	WB	-11.5	-4.6	2.7	7.7	12.6	17.5	18.2	17.7	13.5	6.4	-1.4	-7.2	(29)	
		MCDB	-10.9	-2.8	7.4	15.0	22.5	25.7	24.4	24.4	19.5	10.9	0.0	-6.5	(30)	
	5%	WB	-14.3	-7.5	0.9	6.2	11.0	16.2	17.4	16.6	11.9	4.4	-3.3	-9.7	(31)	
		MCDB	-13.8	-6.0	4.9	12.4	19.6	24.0	23.2	22.5	17.1	7.7	-2.0	-9.1	(32)	
	10%	WB	-16.6	-10.2	-1.0	4.5	9.4	15.1	16.6	15.5	10.2	2.6	-5.3	-12.3	(33)	
		MCDB	-16.1	-9.0	2.3	10.2	16.6	22.1	22.3	20.5	14.6	5.1	-4.0	-11.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	18.2	18.3	14.5	15.5	15.7	14.0	13.8	14.8	13.0	13.4	12.7	(35)	
		MCDBR	14.3	17.4	19.5	20.5	23.6	21.8	19.6	19.7	20.7	17.0	12.3	11.9	(36)	
		MCWBR	13.5	15.1	13.9	12.2	13.0	10.7	9.6	10.8	12.4	11.4	10.3	11.1	(37)	
	5% WB	MCDBR	14.1	16.9	18.0	18.7	21.3	18.0	15.0	17.0	19.3	16.2	11.3	11.7	(38)	
		MCWBR	13.3	14.7	12.8	11.5	12.1	8.9	7.9	9.4	11.6	11.1	9.5	10.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.257	0.287	0.329	0.364	0.377	0.370	0.373	0.359	0.325	0.311	0.289	0.261	(40)	
	taud		2.137	2.106	2.087	2.244	2.286	2.412	2.429	2.434	2.478	2.410	2.207	2.090	(41)	
	Ebn at Noon		729	812	847	867	873	883	873	867	862	793	677	635	(42)	
	Edh at Noon		70	99	125	122	124	110	107	101	85	74	66	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.27	2.33	3.86	4.96	5.36	5.95	5.48	4.65	3.40	2.12	1.18	0.90	(44)	
	RadStd		0.07	0.12	0.15	0.24	0.42	0.49	0.30	0.34	0.40	0.16	0.12	0.06	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page